MILITARY SPECIFICATION

PERISCOPES, TANK: 11747126 and 11747127

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers two types of infrared binocular instruments known as Periscope, Tank: 11747126 and Periscope, Tank: 11747127. Both are similar except that Periscope 11747127 has an extended body.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified.

SPECIFICATIONS

MILITARY

MIL-F-13926	Fire Control Material: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

DRAWINGS

US. Army, Frankford Arsenal

F7297923	Adapter, Pressure Tester (Periscope:11747127)
F7560062	Adapter, Pressure Tester (Periscope:11747126)
F7560189	Fixture, Final Inspection
A7561204	30 KV D.C. Power Supply W/E
F7680673	Fixture, Pressure Testing

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F11747126	Periscope, Tank: 11747126
F11747127	Periscope, Tank: 11747127

PACKAGING DATA SHEET

PS 11747126	Packaging of Periscope, Tank:11747126
PS 11747127	Packaging of Periscope, Tank:11174127

3. REQUIREMENTS

3.1 Fabrication.- The periscopes shall be manufactured in accordance with Drawing F11747126 for Periscope, 11747126; or Drawing F11747127 for Periscope 11747127, and drawings pertaining thereto.

3.2 General specifications.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of specification MIL-F-13926:

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Electrical and electronic assemblies
e. Part identification and marking
f. Workmanship

3.3 Storage temperature.- The periscope shall show no evidence of physical failure after exposure to -70°F (-56.7°C) and +160°F (71.1°C). Each exposure shall be for a minimum of three hours. Subsequent to both exposures the periscope shall meet the requirements of this specification while stabilized at a temperature between +60°F (15.6°C) and +90°F (32.2°C).

3.4 Vibration.- Prior to vibration the periscope shall meet the requirements of 3.10.3. The periscope shall then be vibrated in a vertical plane at a constant frequency of 30 cycles per second, with an amplitude of 0.0625 inch (1.59 mm) [0.125 inch (3.17mm) total excursion], for a period of 5 minutes ± 15 seconds.

3.5 Post-vibration.- Subsequent to vibration the periscope shall show no evidence of physical failure. The periscope shall again meet the requirements of 3.10.3, except that the divergence value shall be within 10 minutes, and the dipvergence value shall be within 5 minutes of the values established prior to vibration (see 3.4).

3.6 Sealing.- The periscope, without head assembly, shall show no evidence of leakage in excess of 0.10 psig (0.007 kg/cm^2) over a period of at least 5.0 minutes when subjected to an external pressure of 5.0 + 0.25 psig (0.351 + 0.017 kg/cm^2).

3.7 Rain.- There shall be no evidence of water within the cavities at the top of the body assembly, just below the head assembly of the periscope, after being subjected to a simulated rainfall for a two hour period. The direction of the rainfall shall be variable from the vertical to 45 degrees from the vertical, and so directed as to equally strike all surfaces of the head assembly (top and four sides) that are normally subjected to rain.

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3.8 Cleanliness

3.8.1 Defects.- There shall be no evidence of moisture, grease, fingerprints, condensates, fractores and adhesive separations on any glass component when viewing through either end of the periscope.

3.8.2 Dirt.- When viewed through the eyepieces of the periscope, with the image tubes energized, the following requirements shall be met. As viewed through the central one-half diameter of the apparent field of view, the optical elements in the periscope shall show no more than 5 particles of dirt one millimeter apparent diameter. There shall be no more than 10 particles of dirt 1.5 mils maximum apparent diameter in the remainder of the field.

3.9 Electrical

3.9.1 Ground potential.- The external unpainted surfaces of the periscope, with the exception of the head assembly, shall be at ground potential at all times.

3.9.2 Dielectric strength.- There shall be no continuous ,or intermittent, arcing or evidence of insulation breakdown when a potential of 17,500 plus 500 volts dc is applied between the inner conductor and ground of the power receptable for a time period of 7.0 seconds plus 3.0 seconds. This requirement shall be met with the image converter tubes installed in the periscope, and the ground leads connected.

3.10 Performance.- The following requirements shall be met with a potential of 16,000 $\pm$ 300 volts DC applied to the power receptable of the periscope. These requirements shall be met subsequent to final sealing of the periscope.

3.10.1 Eyepiece focus.- The eyepiece shall be set between minus 0.75 and minus 1.0 diopter when the image of a target on the flourescent screen is in best focus at the center of the screen.

3.10.2 Resolution.- The resolution on axis of each optical system of the periscope shall be 4.0 minutes of arc or less.

3.10.3 Collimation.- With the rays from an object point 18 to 20 yards (16.46 to 18.29 meters) distant projected into the two objective lenses, simultaneously, the conjugate rays emerging from the two eyepieces shall be divergent in a horizontal plane by 3 degrees 11 minutes $\pm$ 40 minutes measured at the eyepiece, at an interpupillary distance of 65 millimeters. Dipvergence of the target shall not exceed 15 minutes of arc.

3.10.4 Collimation change due to electron image shift.- The periscope shall be subjected to ten cycles of "ON" and "OFF" at the specified voltage (see Par. 3.10). The voltage shall be applied for 15 seconds and removed for 10 seconds. Collimation of the periscopes at the expiration of each "ON" period shall not exceed the requirements of 3.10.3.

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3.10.5 Paraxial magnification.- The paraxial magnification of the two optical systems shall not differ by more than six percent. The paraxial magnification shall be measured at the same points in the field in each eyepiece, i.e., at the edges of the 0.161 inch (4.09mm) diameter field of the eyepiece.

3.10.6 Image tilt.- The image of a plumb line formed by each of the two optical systems shall be vertical within two degrees of arc and shall be parallel to each other within one degree.

3.10.2 Image steadiness.- There shall be no visible flicker, no movement of the image, nor audible sparking when the image of each optical system is observed for at least one minute after the expiration of fifteen seconds tube stabilization time.

3.11 Reliability.- The periscope shall have a mean time between failure (MBTF) of 2000 hours.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three periscopes plus three each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

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4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.3.2.2 Periscope.- Examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II, III and IV herein. Examination and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I, II, III and IV shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I - CLASSIFICATION OF DEFECTS

CLASS
Critical; NONE
MAJOR: AQL 1.0% Defective

NO.	CHARACTERISTIC	REQUIREMENTS	TEST PROCEDURE
101.	Collimation, pre-vibration	3.4	4.6.2
102.	Vibration	3.4	4.6.3
103.	Collimation, post vibration	3.5	4.6.4
104.	Physical failure, post vibration	3.5	4.6.5
105.	Sealing	3.6	4.6.6
106.	Cleanliness	3.8	4.6.7
107.	Eyepiece focus	3.10.1	4.6.9.1
108.	Resolution	3.10.2	4.6.9.2
109.	Collimation	3.10.3	4.6.9.3
110.	Collimation change due to electron image shift	3.10.4	4.6.9.4
111.	Paraxial imagnification	3.10.5	4.6.9.5

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112.	Image tilt	3.10.6	4.6.9.5
113.	Image steadiness	3.10.7	4.6.9.7

MINOR: NONE

NOTE: The tests in Table I shall be conducted at a temperature between +60° and +90°F. (15.6° and 32.2°C).

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- One periscope shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table II.

TABLE II - CLASSIFICATION OF DEFECTS

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.1	Applicable drawings and specifications-visual
302.	General specification	3.2	MIL-F-13926 - visual

4.4.2 Electrical.- Electrical examination and tests shall be conducted prior to final sealing of the periscope. Three periscopes, of each model being procured, shall be selected at random from each 50 produced, or from each month's production, whichever occurs first. Each sample shall meet the requirements and tests in Table III.

TABLE III - CLASSIFICATION OF DEFECTS

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303.	Ground potential	3.9.1	4.6.8.1
304.	Dielectric strength	3.9.2	4.6.8.2

NOTE: The tests in Table III shall be conducted at a temperature between +60° and +90°F (15.6° and 32.2°C).

4.4.3 Environmental.- Three periscopes of each model being procured shall be selected at random from each 50 produced, or from each month's production, whichever occurs first. Each sample that meets the requirement and test for Characteristic 305 shall subsequently meet the requirements and test in Table I.

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TABLE IV - CLASSIFICATION OF DEFECTS

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
305.	Storage Temperature	3.3	4.6.1
306.	Rain	3.7	4.6.7

NOTE: The test for Characteristic 305 shall be conducted at -70°F (-56.7°C) and + 160°F (71.1°C). The test for Characteristic 306 shall be conducted at a temperature between +60° and +90°F (15.6° and 32.2°C).

4.4.4 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specification or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the technical data package (TDP).

4.5.2.2 Contractor design.- In the absence of Government inspection equipment designs, the contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerance used in the design of test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.6 Test methods and procedures

4.6.1 Storage temperature.- The testing equipment utilized in this test shall be in accordance with the "Test facilities" requirements of MIL-F-13926 and the conditions of 3.3. The periscope shall be placed in the test chamber and the temperature of the chamber reduced gradually

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(see Note below) to -70°F (-56.7°C). the periscope shall remain at this temperature for a minimum of 3 hours. At the completion of this 3 hour period, the temperature of the chamber shall be gradually raised (see Note below) to +160°F (71.1°C). The periscope shall remain at this temperature for a minimum of 3 hours. At the completion of this 3 hour period, the temperature of the test chamber shall be gradually reduced to an ambient temperature between +60° to +90°F (15.6°C and 32.2°C). The periscope shall be removed from the test chamber and subjected to the visual, tactile and audible examination of 4.6.5. The periscope shall show no evidence of physical failure. The periscope shall then be subjected to the tests in 4.6.2 through 4.6.9.7 to determine overall conformance to 3.3.

4.6.2 Collimation, pre-vibration.- Prior to vibration the periscope shall be subjected to the test in 4.6.9.3. Collimation of the periscope shall conform to the requirements of 3.10.3. Record the pre-vibration collimation values obtained for evaluation against the post vibration collimation values to determine conformance to the requirements of 3.5.

4.6.3 Vibration.- The periscope shall be subjected to the vibration test utilizing standard vibrating equipment capable of meeting the conditions of 3.4. The accuracy of the test equipment, and test equipment measurements, shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The periscope shall be clamped on the vibrating equipment in its normal mounting, or operating, position by means of an appropriate adapter. The periscope shall be vibrated in the vertical plane for the time duration specified in 3.4.

4.6.4 Collimation, post vibration.- Immediately following vibration the periscope shall again be subjected to the test in 4.6.9.3. Collimation of the periscope shall conform to the requirements of 3.10.3. Record the post vibration collimation values. Any change in collimation, which is evidenced as a difference between the recorded pre-vibration and post vibration values, shall not exceed the divergence and dipvergence tolerances specified in 3.5.

4.6.5 Physical failure, post vibration.- Subesequent to vibration the periscope shall be thoroughly examined for physical defects resulting from poor workmanship and materials to determine compliance with 3.5. A visual examination shall be made for missing or damaged parts. Minimum visual examination shall include elevation lock assembly, cap, chain, nuts, screws, head assembly, and eyepiece assemblies. A tactile examination should also be conducted to determine looseness or improper function of the above items and supporting or assembled members, including the head latching assembly. An audible examination shall be made by holding the periscope close to the ear and vigorously shaking the periscope with both hands while listening for loose internal optical, electronic, and mechanical parts (lenses, prisms, celles, electron tubes, retainers, cables, bonded elements, etc.). Any one of the possible defects defined above shall be cause to consider a periscope defective.

4.6.6 Sealing.- Special testing equipment that conforms to Drawing F7680673, Fixture, Pressure Testing; together with Drawing F7297923, Adapter for Periscope 11747127 or Drawing F7560062, Adapter for Periscope 11747126 shall be utilized for the applicable periscope under test. The

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test procedure for performing the sealing test shall be as detailed in the operating instructions on Drawing F7680673. The periscope shall conform to the sealing requirements of 3.6. The sealing test shall be performed subsequent to the final sealing of the periscope.

4.6.7 Rain test.- This test shall be conducted in accordance with Procedure I of Specification MIL-F-13926. While exposed to the simulated rainfall the periscope shall be secured in its using position, and the high voltage connector shall be sealed against moisture penetration. Immediately following the specified exposure the head assembly shall be removed, and the cavities in the top of the body shall be inspected for moisture penetration. The periscope shall conform to the requirements of 3.7. To facilitate inspection, the external surfaces of the periscope may be dried by compressed air or blotting with water-absorbing material prior to removal of the head assembly. Upon completion of this test the periscope shall be thoroughly dried.

4.6.7 Cleanliness

4.6.7.1 Defects.- This test shall be performed with the periscope deenergized. Utilizing two 15 watt fluorescent lamps, desk or bench tube, view by reflected light the glass components of the head assembly, objective assembly and eyelens assembly. The method of inspection shall be by the technique of shadowing. The periscope shall meet the cleanliness requirements of 3.8.1.

4.6.7.2 Dirt.- This test shall be performed with the periscope energized. Prior to the inspection for dirt the image tubes shall be permitted a stabilization ("warm-up") period of 15 seconds. The screen surface of each image tube shall then be visually inspected through the appropriate eyepiece for evidence of dirt. The periscope shall meet the cleanliness requirements of 3.8.2. "Cold" and "hot" spots on the screen surface of the image tube shall not be considered dirt.

4.6.8 Electrical.- Electrical examinations and tests shall be conducted prior to final sealing of the periscope.

4.6.8.1 Ground potential.- Standard measuring equipment shall be utilized for this test. The periscope shall conform to the requirements of 3.9.1.

4.6.8.2 Dielectric strength.- Conduct this test in accordance with Method 301 of Standard MIL-STD-202, with the image converter tubes installed in the periscope, and the ground leads connected. A high-voltage power source that conforms to Drawing A7561204 shall be utilized to supply the voltage for this test. During test voltage application a visual and audible examination shall be conducted on the periscope circuits for evidence of grounding. Subsequent to voltage application the periscope circuits shall be subjected to a visual examination for evidence of insulation breakdown and damage due to corona. The periscope shall conform to the requirements of 3.9.2.

4.6.9 Performance.- Special testing equipment that conforms to Drawing F7460189 shall be utilized in testing the performance characteristics of the periscope. The voltage applied to energize the periscope shall conform to the requirements of 3.10, and shall be supplied by a high-voltage power source

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that conforms to Drawing A7561204. The high-voltage power source shall have a 2 minute warm-up period prior to applying the operating voltage to the periscope. All performance tests shall be conducted subsequent to final sealing of the periscope.

4.6.9.1 Eyepiece focus.- A calibrated dioptometer having a magnification of at least three power (3X) shall be utilized for measuring the eyepiece focus of each optical system in the periscope. Set the dioptometer diopter scale to zero, and adjust the dioptometer eyepiece for best focus of the dioptometer reticle. Position the dioptometer at each eyepiece of the periscope, in turn, and obtain best focus of the phospor grain on the screen of the image tube by adjustment of the dioptometer diopter scale. The phosphor grain on the screen of the image tube shall be in best focus, in each optical system, within the diopter tolerance specified in 3.10.1, as measured on the dioptometer diopter scale.

4.6.9.2 Resolution.- Perform this test in sequence to the test for eyepiece focus. Focus the dioptometer by means of the diopter scale adjustment, and obtain correct line count of the resolution target image in all four meridians. Resolution of the target image, in all four meridians, shall be obtained within 1/8 diopter of the measured eyepiece focus value obtained for the eyepiece in the periscope. The resolution of each optical system in the periscope shall conform to the requirements of 3.10.2.

4.6.9.3 Collimation.- Perform this test with the lower collimators in position at the periscope eyepieces.

4.6.9.3.1 Divergence.- Divergence of 3 degrees 11 minutes is "built into" the lower collimator assembly of the testing equipment when it is calibrated in accordance with the set-up instructions on Drawing F7560189. The amount of divergence greater than, or less than, 3 degrees 11 minutes shall be measured by means of the horizontal scale divisions on the center of the reticle in the lower collimator, with respect to the vertical reticle line in the upper collimator. The scale divisions represent 10 minutes of arc, for a total of 40 minutes of arc on each side of the vertical reticle line. Adjust the lower gimbal mounting to obtain coincidence of the centers of the lower and upper right collimator reticle patterns. Read the amount of divergence greater than, or less than, the established 3 degrees 11 minutes on the reticle scale in the lower left collimator. The conjugate rays emerging from the two eyepieces shall be divergent in a horizontal plane within the tolerance specified in 3.10.3. Record divergence value.

4.6.9.3.2 Dipvergence.- Dipvergence of zero minutes is "built into" the lower collimator assembly of the testing equipment when it is calibrated in accordance with the set-up instructions on Drawing F7560189. Dipvergence shall be measured by means of the center scale divisions on the vertical reticle line in the lower collimator, with respect to the horizontal reticle line in the upper collimator. The scale divisions represent 5 minutes of arc, for a total of 15 minutes of arc on each side of the horizontal reticle line. Adjust the lower gimbal mounting as in 4.6.9.3.1 and read the amount of dipvergence on the reticle scale in the lower left collimator. The conjugate rays emerging from the two eyepieces shall not exceed the dipvergence tolerance specified in 3.10.3. Record dipvergence value.

4.6.9.4 Collimation change due to electron image shift.- This test shall require a suitable "off-on" cycling switch to be inserted in the high voltage line between the high voltage power supply and the periscope. The

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power supply unit shall not be cycled for this test. Adjust the lower gimbal mounting as in 4.6.9.3.1. Subject the periscope to the "on-off" voltage cycling specified in 3.10.4, while directing a line of sight through the lower collimators. At the expiration of each "on" period the collimation of the periscope shall conform to the requirements of 3.10.4, as measured on the lower collimator reticles with reference to the upper collimator reticles.

4.6.9.5 Paraxial magnification.- the bow tie reticle pattern in the upper collimators, in conjunction with the upper vertical reticle scale in the lower collimators shall be utilized for measuring magnification of the periscope. Adjust the lower gimbal mounting as in 4.6.9.3.1. Record the point of intersection of the bow tie reticle pattern image with the vertical reticle scale image as the magnification value for the periscope's right optical system. Aline the lower left collimator with the upper collimator, as above, and obtain the magnification value of the periscope's left optical system. Each division on the vertical reticle scale represents 2 percent. Any difference in magnification between the left and right optical systems shall not exceed the tolerance specified in 3.10.5.

4.6.9.6 Image tilt.- The horizontal scale at the bottom of the reticle in the lower collimators, in conjunction with the vertical reticle line in the upper collimators, shall be utilized for measuring image tilt in each optical system of the periscope. Each division on the horizontal reticle scale represents 30 minutes of arc. Adjust the lower gimbal mounting as in 4.6.9.3.1. Record the point of intersection of the vertical reticle line image with the horizontal scale image as the image tilt value for the right optical system. Aline the lower left collimator with the upper collimator, as above, and record the image tilt value for the left optical system. Any image tilt in the optical systems of the periscope shall not exceed the tolerances specified in 3.10.6.

4.6.9.7 Image steadiness.- This test shall consist of a simultaneous visual examination of the image screens and an audio examination of the periscope when it is energized for the specified time. The periscope shall conform to the requirements of 3.10.7.

4.6.10 Reliability demonstration test.- A reliability demonstration test shall be conducted as part of first article test to assure compliance with the requirement of paragraph 3.11 at a 70% confidence level. A minimum of 25 hours on each sample periscope assembly (sample size 3) shall be accumulated after exposure to each nonoperating environment of this specification. Additional hours are required at 75 ±15°F in order to accumulate a minimum of 800 hours on each sample. The sample periscope assembly shall be tested until the total operating time (2408 hours minimum) satisfies the following relationship or until 6 failures have occurred or 14,000 hours (maximum) have been accomplished.

$$2000 \geq \frac{2T}{\text{Chi Square } .30,\ 2(r+1)}$$

T = Total test hours

Chi Square .30, 2(r+1) = Chi Square lower 70% confidence level with 2 (r+1) degrees of freedom

r = Number of chargeable failures

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4.6.10.1 Reliability data.- Data collected will consist of all equipment malfunctions, number of hours completed when these malfunctions occurred, nature and cause of malfunction, time required to correct malfunction and required corrective action.

4.6.10.2 Reliability failure definition.- Any malfunction which renders the periscope assembly inoperative or operative outside the limits of this specification. In determination of chargeable failure, the following clarification shall apply:

a. If, upon failure, analysis (to determine the cause of malfunction), an assignable cause (related to manufacturing process, procedure, basic design weakness, etc.) is determined as the basis of the malfunction, the malfunction will not be chargeable as a reliability failure; however, the correction to the manufacturing process or procedure (et al) shall be accomplished and the testing resumed at the environment at which the malfunction occured.

b. If upon failure analysis no assignable cause is evident and the malfunction is considered random in nature, the malfunction shall be a "chargeable" reliability failure and, as such, must be considered in the above demonstration test equation (4.6.10).

5. PREPARATION FOR DELIVERY

5.1 Packing, packing and marking.- Packaging, packing and marking shall be in accordance with Packaging Data Sheet 11747126 for Periscope 11747126 and Packaging Data Sheet 11747127 for Periscope 11747127. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The infrared periscopes are used in conjunction with the vehicle infrared headlamps to facilitate night driving of tanks, self-propelled vehicles and armored personnel carriers.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Selection of applicable levels of preservation, packaging and packing.
c. Applicable packaging data sheet (see 5.1).
d. Applicable stock number.
e. Provisions for First Article Testing.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 6650-A090

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